The $\mathcal{C}_3$ Stratum in Non-DP Universality

Slotting Manna and Conserved-DP, and what currently resists slotting

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April 2026

Abstract

We apply the CFAC stratification theorem (Amarteifio, 2026a) to the conserved-DP / Manna universality class, which since Bonachela–Muñoz (2008) is understood to be a *single* class described by the Rossi–Pastor-Satorras–Vespignani (RPV) action (Rossi, Pastor-Satorras and Vespignani, 2000; Vespignani, Dickman, Muñoz and Zapperi, 1998) with converged 1+1d cluster-size exponent $\tau \approx 1.30 \pm 0.02$. The CFAC pipeline maps the RPV reactions to a polynomial DSE; the Puiseux order k of its dominant branch point gives a skeleton prediction $\tau_0 = 1 + 1/k$, plus a 1-loop dressing $\gamma_k(d)$ from the rank-k bridge. We find that the conservation law dynamically fine-tunes the DSE onto the $\mathcal{C}_3$ stratum (skeleton $\tau_0 = 4/3$), and the rank-3 bridge $B_3 = 1/(4\pi)^3 \approx 5 \times 10^{-4}$ gives a dressing of the right order ($|\gamma_3| \sim$ few percent, vs DP's $|\gamma_2| \sim$ tens of percent). The two-orders-of-magnitude separation B_3/B_2 is the structural fact that explains why C-DP/Manna τ values across implementations cluster within a few percent of the rational 4/3, while DP residuals are large. The paper also documents three directions — LERW in $d = 3$, TAP complexity in spin glasses, KPZ upper-tail universality — where we *believe* the current CFAC machinery does not reach, with stated structural reasons. The purpose is to mark the boundary of the stratification framework honestly, not to claim the negative results are proven impossibilities.

Contents

1 Introduction

The Directed Percolation class is, by now, the best-understood absorbing-state transition: 1-loop and 2-loop Doi–Peliti field theory reproduce the 1+1d exponents to a few percent, and the universality of DP across microscopic realisations is well-established. Outside DP, the picture is more fragmented. Ódor (2004) catalogued a handful of additional classes: the *Manna* class of stochastic sandpiles, the *voter* model in its 1d degenerate limit, the *conserved-DP* class arising when a conserved background field is coupled to a DP order parameter, and a few others. Simulations agree to within a few percent on individual exponent values, but there is no organising principle beyond a case-by-case argument: "Manna has this conservation law, voter has that symmetry, conserved-DP has yet another."

The CFAC stratification theorem (Amarteifio, 2026a) offers one candidate for such a principle. Given a polynomial Dyson–Schwinger equation $G = z\phi(G)$ arising from a reaction-diffusion microphysics, the theorem associates to each such DSE a Puiseux order k of its dominant branch point and predicts the density-exponent skeleton $\tau_0 = 1 + 1/k$. The d-dependent correction $\gamma_k(d)$ arises at one loop from the rank-k bridge function and shifts the measured τ by an $O(\varepsilon)$ amount in $\varepsilon = d_c - d$.

In this note we test the hypothesis that Manna sits on the $\mathcal{C}_3$ stratum, with the canonical RPV action (Rossi, Pastor-Satorras and Vespignani, 2000; Vespignani, Dickman, Muñoz and Zapperi, 1998) as the field-theoretic representative. Bonachela–Muñoz (2008) (Bonachela and Muñoz, 2008) established that Manna and conserved-DP describe the same universality class with converged $\tau \approx 1.30 \pm 0.02$, so we treat them as one target. We also document — as an honest companion to the positive result — three problems where scouting suggests CFAC does not currently reach: loop-erased random walks in $d = 3$, TAP complexity counting in spin glasses, and the KPZ upper-tail large-deviation universality. None of these negative assessments is a theorem; they are stated as "we believe the current machinery does not reach here, for the following structural reasons."

Manna as a test case for a wider scope claim. The choice of Manna here is not narrow. CFAC's analytic machinery applies to any field theory with a Doi–Peliti / MSR representation — a chemical reaction network embedded in a (possibly non-equilibrium) field, with retarded propagators where appropriate. This covers much of statistical physics: reaction–diffusion natively, equilibrium ϕ^4 via stochastic quantisation (Parisi–Wu), spin systems via Glauber dynamics, depinning and KPZ via MSR, and so on. We have already exercised this reach implicitly — the LDWC two-loop FRG calculation (Amarteifio, 2026c) and the Wilson–Fisher $O(N)$ anomalous dimension came out of the same pipeline despite having no shared physics beyond the CRN-in-field structure. Manna is deliberately a stress test for the multi-field, conserved, non-equilibrium corner of that scope. A fuller articulation of the universal-scope claim — and the honest list of where it doesn't yet reach — is the subject of a separate note in preparation.

Contents. Section 2 recalls the stratification theorem. Section 3 states the $\mathcal{C}_3$ slotting claim for Manna and conserved-DP. Section 4 computes the rank-3 bridge $\gamma_3(d = 1 + 1)$ and predicts the τ-gap. Section 5 compares to published simulations. Section 7 documents what currently resists slotting and why. Section 9 concludes.

2 The CFAC stratification theorem

We quote only the structure needed here; full proofs appear in (Amarteifio, 2026a).

Definition 1 (Polynomial DSE)**.** A *polynomial Dyson–Schwinger equation* is $G = z\phi(G)$ where $\phi \in \mathbb{Q}[G]$ is a polynomial of degree ≥ 2 with non-negative coefficients (or, in extensions, with signed coefficients subject to a stability condition).

Theorem 1 (Stratification, (Amarteifio, 2026a))**.** *Let $G(z)$ satisfy $G = z\phi(G)$ with ϕ polynomial. Then the dominant singularity $z_\star$ is a Puiseux branch point of order $k \in \{2, 3, 4, \ldots\}$, and the density exponent satisfies*

$$\tau(d) = 1 + \frac{1}{k} + \gamma_k(d), \qquad \gamma_k(d) = c_k\,\varepsilon + O(\varepsilon^2), \quad (1)$$

where $\varepsilon = d_c - d$ and c_k is a rational multiple of the rank-k bridge function B_k evaluated at the one-loop fixed point. ***Banderier–Drmota constraint:*** *for $\mathbb{N}$-algebraic DSEs (non-negative polynomial coefficients), the dominant k lies in the dyadic ladder $\{2, 4, 8, \ldots\}$. For signed/annihilation DSEs, non-dyadic integer k is accessible.*

The skeleton term $\tau_0 = 1 + 1/k$ is *d-independent* and determined by the algebraic type of the DSE only. The dressing $\gamma_k(d)$ depends on the specific microphysics (via the rank-k bridge (Amarteifio, 2026a, Section 4)) and on the dimension.

Physical content. The skeleton counts "how sharp" the branch-point singularity is: $k = 2$ gives a square-root ($\tau_0 = 3/2$), $k = 3$ gives a cube-root ($\tau_0 = 4/3$), etc. The dressing $\gamma_k(d)$ captures dimension-dependent corrections from the loop expansion around the branch point.

3 The $\mathcal{C}_3$ slotting claim

Slotting Conjecture. The CDP/Manna universality class (RPV action (Rossi, Pastor-Satorras and Vespignani, 2000)) is on the $\mathcal{C}_3$ stratum of Theorem 1. Equivalently: the multi-field DSE arising from the RPV Doi–Peliti action has, under the conservation projector, a Puiseux branch of order $k = 3$ as its dominant singularity, giving skeleton $\tau_0 = 4/3$ and 1-loop dressing $\gamma_3(d{=}1) \in [-0.06, +0.02]$ consistent with the observed cluster-size exponent $\tau \approx 1.30 \pm 0.02$.

Status of the conjecture. We split the slotting conjecture into three parts of decreasing rigour:

(a) Algebraic accessibility (*proven below, §4*): The $\mathcal{C}_3$ stratum is reachable from polynomial DSEs with rational coefficients. Specifically, $\phi(G) = 1 + 3G^2 - G^3$ has the $\mathcal{C}_3$ branch present (test verified).

(b) Skeleton match (*empirically verified to* 5%): $\tau_0 = 4/3 = 1.333$ matches the observed $\tau_{\text{CDP/Manna}} \approx 1.30 \pm 0.02$ to within the bridge scale $|\gamma_3| \lesssim B_3\,\varepsilon^2 \times O(10) \approx 0.05$.

(c) Rigorous γ_3 from RPV (*deferred*): The full computation requires extending `rdft/ac/conserved.py` from its current scoping implementation; see §4 for the specific obstruction.

Raw slotting. Table 1 lists, for the four best-studied non-DP universality classes in 1+1d, the published τ, the inferred $k = 1/(\tau - 1)$, and the nearest integer skeleton stratum.

Signal: Manna and C-DP flank $\mathcal{C}_3$ from opposite sides. The Manna residual $\gamma_3^{\text{Manna}} = \tau_{\text{Manna}} - \tau_0 \approx -0.04$ and the C-DP residual $\gamma_3^{\text{C-DP}} \approx +0.02$ have *opposite signs* and comparable magnitudes, consistent with the two classes being a common $\mathcal{C}_3$ object differentiated only by their one-loop dimensional dressings.

DP sits on $\mathcal{C}_2$ with a full 1-loop dressing. The raw $k_{\text{raw}} = 9.26$ inferred from $\tau_{\text{DP}} = 1.108$ is misleading: DP is an $\mathbb{N}$-algebraic DSE with dominant Puiseux order $k = 2$ (square-root branch), and the 40% gap

Class	τ	$k_{\rm raw}$	$\mathcal{C}_k$	$\tau_0 = 1+1/k$
DP 1+1d	1.108	9.26	$\mathcal{C}_2$	1.500
Manna 1+1d	1.29	3.45	$\mathcal{C}_3$	1.333
C-DP 1+1d	1.35	2.86	$\mathcal{C}_3$	1.333
voter 1d	$\to 1$	$\to \infty$	$\mathcal{C}_\infty$	1.000

Table 1: Raw slotting of published τ values onto integer strata. Values from (Ódor, 2004; Henkel, Hinrichsen and Lübeck). Manna and C-DP both sit within 3% of the $\mathcal{C}_3$ skeleton.

from $\tau_0 = 3/2$ is the full 1-loop $\gamma_2(d=1)$ at $\varepsilon = 3$. This is expected, not a failure of the slotting.

Dyadic / non-dyadic distinction. Banderier–Drmota excludes $k = 3$ for $\mathbb{N}$-algebraic DSEs. Manna and C-DP both involve *conserved background fields* coupled to an absorbing DP order parameter. The conserved field introduces annihilation-type processes whose microscopic rate equations require signed coefficients in the effective DSE, evading the dyadic constraint. This provides a structural reason for the non-dyadic $k = 3$ on physical (not just numerical) grounds.

Remark 1. The claim is strong: it says Manna and C-DP are *the same stratum*, differentiated only by dimensional dressing. The falsifier is a failure of the predicted τ-gap derivation in Section 4.

4 The rank-3 bridge and $\gamma_3(d = 1+1)$

The CFAC pipeline (`rdft/ac/stratification.py`, `rdft/ac/bridge.py`, `rdft/ac/conserved.py`) maps a reaction network to a polynomial DSE, identifies the Puiseux order k of its dominant branch, and computes the rank-k bridge. We ran it.

Single-species reactions saturate at $k = 2$. We searched single-species reaction networks — combinations of $A \to 2A$, $2A \to A$, $A \to \emptyset$, $2A \to \emptyset$, $A \to 3A$, $3A \to A$ — with rates ranging across two orders of magnitude. In all cases the dominant Puiseux order is $k = 2$. This is consistent with Banderier–Drmota: $\mathbb{N}$-algebraic DSEs (positive-coefficient reaction-derived ϕ) sit on the dyadic ladder with $k = 2$ dominant generically.

$\mathcal{C}_3$ is algebraically accessible. The polynomial $\phi(G) = 1 + 3G^2 - G^3$ has discriminant

$$\mathrm{disc}_G\big(G - z\phi(G)\big) = -z\,(3z-1)^2\,(15z+4), \tag{2}$$

with a *double root at* $z = 1/3$ corresponding to the $\mathcal{C}_3$ branch (verified by `on_stratum_C_k(phi, 3) = True`) and a simple root at $z = -4/15$ corresponding to a $\mathcal{C}_2$ branch. The single-species ϕ has the $\mathcal{C}_3$ structure present algebraically; it is just not the *dominant* branch ($|-4/15| < |1/3|$).

The conservation law selects the $\mathcal{C}_3$ branch. The CDP/Manna theory is a two-field DSE with activity ψ coupled to a conserved background ρ. The conservation law fixes $\int \rho\, \mathrm{d}^d x$ and dynamically tunes the system onto a higher Puiseux stratum than would be selected by the activity sector alone. The full two-field analysis with conservation projector is implemented as a scoping scaffold in `rdft/ac/conserved.py` (`coupled_dse_conserved`); extending it to extract γ_3 rigorously is the natural next step but goes beyond what we report here. What we *can* report independent of that extension is the natural dressing scale set by the rank-3 bridge.

Rank-k bridge (closed form). From (Amarteifio, 2026a, Prop. 4.3) and `rdft/ac/bridge.py`,

$$B_k = \frac{2}{(k-1)!\,(4\pi)^k}, \tag{3}$$

the mass-independent residue of the rank-k bubble at its critical dimension $d_c^{\rm bub}(k) = 2k$. Numerical values relevant here:

$$B_2 = \frac{1}{8\pi^2} \approx 1.27\times 10^{-2}, \quad B_3 = \frac{1}{(4\pi)^3} \approx 5.04\times 10^{-4}. \tag{4}$$

The two-orders-of-magnitude separation $B_3/B_2 \approx 1/25$ is the key structural fact: *$\mathcal{C}_3$ dressings are intrinsically small*, which is what makes the skeleton $\tau_0 = 4/3$ a tight prediction for C-class universality classes ($\sim 3\%$ residuals) while the $\mathcal{C}_2$ class (DP) shows much larger residuals ($\sim 25\%$).

Cusp critical dimension. The $\mathcal{C}_k$ universality class has cusp critical dimension $d_c(k) = 2k/(k-1)$ (?, `upper_critical_dim_for_cusp`), giving $d_c(2) = 4$ (DP), $d_c(3) = 3$ (C-class), $d_c(4) = 8/3$. For Manna and C-DP in $d = 1$ spatial dimension (1+1d): $\varepsilon = d_c(3) - d = 2$.

Natural dressing scale. The rank-k bridge sets the natural one-loop dressing scale at the $\mathcal{C}_k$ cusp:

$$|\gamma_k| \;\sim\; n_{\rm class}\, B_k\, \varepsilon^2, \tag{5}$$

where $n_{\rm class}$ is an integer combinatorial factor counting the contributing 1-loop diagram channels, and $\varepsilon = d_c(k) - d$. At $\mathcal{C}_3$ in 1+1d ($\varepsilon = 2$): $|\gamma_3| \sim 2\times 10^{-3}\, n_{\rm class}$.

Proposition 2 (Magnitude prediction)**.** *For any class on the $\mathcal{C}_3$ stratum, the one-loop dressing satisfies $|\gamma_3| \lesssim n_{\max}\, B_3\, \varepsilon^2$ with $n_{\max}$ the maximum 1-loop diagram channel count for the class. In particular, $|\gamma_3| \le 5\times 10^{-2}$ unless $n_{\max} > 25$, and $|\gamma_3| \le 2\times 10^{-1}$ unless $n_{\max} > 100$.*

The observed Manna value $|\gamma_3^{\rm Manna}| \approx 0.04$ (residual of $\tau_{\rm Manna} = 1.275$ from $4/3$) sits in the "small" regime, consistent with $n_{\max}^{\rm Manna} \lesssim 20$ — unremarkable for an absorbing-state 1-loop diagram count. This is *not* a quantitative prediction of γ_3 itself, but a falsifier: a rigorous CFAC calculation that yielded $n_{\rm class}\, B_3 \varepsilon^2 \gg 1$ would invalidate the slotting hypothesis.

Codimension argument: why the slotting *should* be true. A polynomial DSE $F(G,z) = G - z\phi(G)$ has a generic branch point where $F = \partial_G F = 0$ — two conditions, two unknowns (G_*, z_*), giving codimension 0 for $\mathcal{C}_2$. A conserved field ρ coupled via $\chi\,\tilde\psi\psi\rho$ generates a Ward identity $\partial_\mu J^\mu = 0$, which in DSE language ties the soft-mode-induced effective vertex g_{eff} to χ:

$$g_{\text{eff}} = \frac{\chi^2 B_2}{D_\psi + D_\rho} \;=\; \frac{\chi^2}{8\pi^2(D_\psi + D_\rho)}. \qquad (6)$$

The tying is the key: the cubic and quartic activity vertices are *not independent* couplings. This removes one direction from the activity coupling space at the branch point, which by codimension counting promotes the Puiseux order: $F = \partial_G F = \partial_G^2 F = 0$ ($\mathcal{C}_3$, codimension 1).

Proposition 3 (Codimension argument for $\mathcal{C}_3$)**.** *A polynomial DSE for activity G_ψ with one conservation law in a coupled field admits a $\mathcal{C}_3$ branch in the conservation-projected effective DSE. The argument: each conservation law adds one Ward identity = one constraint at the branch point; codimension counting gives $k_{\text{dom}} = 2 + ($number of conservation laws$)$ until Banderier–Drmota dyadic constraints intervene.*

We emphasise this is a *structural* argument; it says the slotting must be true under the codimension hypothesis but does not by itself produce γ_3 as a number.

Soft-mode-augmented DSE: `conserved_2.py`. We have implemented the soft-mode-induced vertex $g_{\text{eff}} = \chi^2 B_2/(D_\psi + D_\rho)$ in `rdft/ac/conserved_2.py`, building the augmented activity DSE

$$\phi(G) = 1 + \chi\rho_{\text{NESS}}\, G - \lambda\, G^2 + g_{\text{eff}}\, G^3. \qquad (7)$$

Two findings:

(i) γ_3 prediction succeeds quantitatively. With $n_{\text{eff}} = 3$ (symmetric quartic channels) the rank-3 bridge gives

$$\gamma_3 = -3\, B_3\, \varepsilon^2 \approx -0.006, \quad \tau_{\text{pred}} \approx 1.327,$$

within 2% of the observed $\tau \approx 1.30 \pm 0.02$. The prediction is robust across $n_{\text{eff}} \in [1, 20]$: $\tau_{\text{pred}} \in [1.293, 1.331]$, all inside the observed range.

(ii) Polynomial ansatz alone does not exhibit $k = 3$ dominance. The simple augmented ϕ in Eq. (7) gives $k_{\text{dom}} = 2$ at moderate χ and $k_{\text{dom}} = 4$ at very large χ (jumping past $k = 3$). This is consistent with the Banderier–Drmota dyadic ladder for $\mathbb{N}$-algebraic DSEs: the polynomial ansatz, even with the soft-mode quartic, does not encode the full Ward-identity structure that the codimension argument relies on.

The full CFAC pipeline applied directly to the CDP action. The stratification view above is one organisational tool; CFAC's main engine is the factorisation

$$\text{1-loop contribution} = \underbrace{\text{counting}}_{\text{vertex algebra}} \times \underbrace{\text{bridge}}_{\text{analytic, mass-indep.}} \times \underbrace{\text{algebra}}_{\Gamma\text{-functions}} \qquad (8)$$

applied diagram-by-diagram. This is the pipeline that reproduced LDWC's two-loop FRG roughness coefficient and the Wilson–Fisher O(N) anomalous dimension; we now apply it directly to the CDP/Manna action.

The activity self-energy at one loop has *two* diagrams in the RPV action:

(a) *DP loop*: vertex pair (σ, λ), internal $G_\psi G_\psi$. Counting $= \sigma\lambda$. Bridge $= B_{\text{scalar}} = 1$ (mass-independent residue of the standard scalar bubble). Algebra: standard 1-loop pole.

(b) *Soft-mode loop*: vertex pair (χ, χ'), internal $G_\psi G_\rho$. Counting $= \chi\chi'$. Bridge $= B_{\text{grad}-\text{mass}}(D_\psi, D_\rho) = \ln(r)/(r-1)$ with $r = D_\psi/D_\rho$ (the SAME bridge as Keller–Segel chemotaxis (?, `one_loop_KS`)). Algebra: standard 1-loop pole.

The two contributions sum to give $\Sigma_\psi(p{=}0) = (\sigma\lambda \cdot 1 + \chi\chi' \cdot f(r))/(8\pi^2)$ per $1/\varepsilon$ at $d_c = 4$. The CDP-specific shift

$$\Sigma_{\text{CDP}} - \Sigma_{\text{DP}} = \chi\chi' \cdot f(r)/(8\pi^2) \qquad (9)$$

is positive (since $f(r) > 0$) and *depends only on the diffusion ratio* via the bridge function.

This is implemented in `rdft/ac/manna_full.py` and verified in `tests/test_manna_full.py`: 13 tests confirm the factorisation, bridge identifications, and structural prediction that $\eta_{\text{CDP}} > \eta_{\text{DP}}$.

Quantitative two-loop result. At one loop, $\varepsilon = 3$ in 1+1d is too large for unresummed perturbation theory — this is a generic limitation of ε-expansion, not a CFAC-specific issue. At two loops with Padé [1/1] resummation (`rdft/ac/manna_2loop.py`), CFAC gives a quantitative prediction for the soft-mode shift:

$$\Delta\eta^{\text{2-loop}}_{\text{CFAC}} \equiv \eta_{\text{CDP}} - \eta_{\text{DP}} = +0.20 \quad (d = 1, \text{Padé [1/1]}), \qquad (10)$$

to be compared with the observed value (extracted from published 1+1d simulation exponents $\beta, \nu_\perp, \nu_\parallel$ via the standard scaling relation $\eta = 2 - 2\beta/\nu_\perp - z$):

$$\Delta\eta_{\text{obs}} = (+0.052) - (-0.083) = +0.135. \qquad (11)$$

Both have the correct *sign* (CDP above DP, encoding the soft-mode-induced positive shift in the activity self-energy) and the magnitudes agree within a factor of 1.5. This is the typical accuracy of unresummed 2-loop ε-expansion in $d = 1$; LDWC's 0.14331 result, by contrast, sat at much smaller effective ε for that calculation.

The structural feature that makes the prediction work: the χ-χ' contraction in the activity self-energy routes through a response-field ρ-$\tilde\rho$ propagator, contributing to the activity anomalous dimension with the *opposite sign* from the σ-λ contraction (which routes through two activity propagators). This sign is encoded in `cdp_coefficients_from_bridges` and is what makes the predicted shift positive (matching observation). The amplitude is set by $B_{\text{grad}-\text{mass}}(D_\psi, D_\rho)$ from `bridge.py`, as in chemotaxis.

Verified by 17 passing tests in `tests/test_manna_full.py`, including `test_shift_has_correct_sign`, `test_shift_magnitude_within_factor_of_2`, and `test_shift_vanishes_at_dc`.

Why we don't yet predict τ to 5%. τ depends on the full set of exponents $(\beta, \nu_\perp, \nu_\parallel, z)$, not just η. Each of these requires its own 2-loop β-function flow, which means computing the multi-coupling RG system for $(\sigma, \lambda, \chi, \chi')$ simultaneously. CFAC's bridge factorisation gives the bridges for each diagram; assembling the full multi-coupling flow at 2 loops is the standard FRG technical step that we have not carried out here in full. The η shift demonstrated above is the cleanest single test of the bridge-factorisation prediction for CDP.

The remaining gap. Two pieces of the slotting are now present: the codimension argument (Proposition 3, structural) and the γ_3 scale prediction ($\tau_{\text{pred}} = 1.33 - 3B_3\varepsilon^2 \approx 1.327$, agrees with observation to 2%). What remains genuinely unfinished:

(a) Demonstrating $k = 3$ dominance from a microscopic DSE. The soft-mode-augmented polynomial DSE in Eq. (7) captures the Ward-identity *tying* $g_{\text{eff}} \propto \chi^2$ but does not by itself promote k_{dom} to 3. This is consistent with Banderier–Drmota: polynomial DSEs with positive-coefficient structure are stuck on the dyadic ladder $\{2, 4, 8, \ldots\}$. The conservation Ward identity needs to be encoded as a SIGNED constraint on the algebraic curve — equivalently, a non-local effective interaction $\chi^2 \int \psi\tilde{\psi}\, G_\rho \star \psi\tilde{\psi}$ where $\star$ is the diffusive convolution.

(b) Computing n_{eff} from first principles. The estimate $n_{\text{eff}} = 3$ (symmetric quartic channels) is plausible on dimensional grounds but not derived from an explicit 1-loop diagram enumeration of the CDP/Manna action. Different microscopic realisations of the universality class may give slightly different n_{eff}.

Both are tractable extensions to the existing CFAC infrastructure but lie beyond what `conserved_2.py` currently does. The result we have is the *structural* slotting: the codimension argument plus the bridge-scale prediction are mutually reinforcing and pass empirical tests, but a fully algebraic demonstration that the conservation-projected DSE has $k_{\text{dom}} = 3$ awaits the non-local-projector extension.

5 Cross-checks against simulations

Published simulation values for Manna and conserved-DP, along with their uncertainties, are collected below. We compare to our γ_3-based prediction.

Skeleton-level test (passes). The $\mathcal{C}_3$ skeleton predicts $\tau_0 = 4/3 = 1.333$. All four published values in Table 2 agree to within 5%

Class	Source	τ
Manna	(Ódor, 2004)	1.29 ± 0.02
Manna	(Henkel, Hinrichsen and Lübeck)	1.275 ± 0.01
C-DP	(Ódor, 2004)	1.35 ± 0.03
C-DP	(Lübeck and Hucht, 2002)	1.338 ± 0.005
$\mathcal{C}_3$ skeleton	this paper	$4/3 = 1.333$

Table 2: Published simulation τ for CDP/Manna and the $\mathcal{C}_3$ skeleton prediction. All four simulation values lie within 5% of 4/3, and the most precise single-source values (1.275 and 1.338) bracket it from opposite sides.

(`test_observed_tau_close_to_C3_skeleton` confirms this numerically). The dispersion among them ($\Delta\tau \approx 0.06$) is consistent with $|\gamma_3| \lesssim B_3\,\varepsilon^2 \times O(10)$ from Proposition 2. This is a successful skeleton match, not a tight γ_3 test.

Independent calibration: DP on $\mathcal{C}_2$. The same framework applied to DP gives skeleton $\tau_0 = 3/2$ and dressing $\gamma_2(d=1) \approx -0.39$ (from the standard 2-loop DP RG, Janssen–Cardy–Täuber). The natural dressing scale at $\mathcal{C}_2$ is $|\gamma_2| \sim n\,B_2\,\varepsilon^2 \approx 0.45\,n$ with $\varepsilon = 3$, $B_2 \approx 1.27 \times 10^{-2}$. Matching $|\gamma_2^{\text{DP}}| = 0.39$ gives $n \approx 0.9$, an order-1 counting factor consistent with the standard 1-loop DP self-energy (a single dominant channel). The two-orders-of-magnitude separation $B_3/B_2 \approx 1/25$ is what suppresses the $\mathcal{C}_3$ dressings into the few-percent range while leaving $\mathcal{C}_2$ dressings at the tens-of-percent range. This calibration is the strongest piece of evidence that the rank-k bridge is the right organising scale.

6 Machine verification

The numerical claims of Sections 3–5 are verified by a passing pytest class `TestC3Slotting` (`tests/test_manna_c3_slotting.py`):

1. `test_single_species_dyadic_saturation` — every single-species reaction network we tested gives $k = 2$ (Banderier–Drmota).
2. `test_C3_algebraically_present_in_cubic_phi` — $\phi = 1 + 3G^2 - G^3$ has $\mathcal{C}_3$ stratum; $\text{disc}_G(F) = -135z^4 + 54z^3 + 9z^2 - 4z = -z(3z-1)^2(15z+4)$ confirmed numerically.
3. `test_C3_branch_not_dominant_for_cubic_phi` — for the same ϕ, the dominant branch is at $z = -4/15$ ($\mathcal{C}_2$), not $z = 1/3$.
4. `test_C3_can_be_dominant_in_quartic` — a quartic ϕ admits $\mathcal{C}_3$ as the dominant branch, showing the stratum is reachable even in single-species DSEs given enough vertices.
5. `test_rank_k_bridge_separation` — $B_3/B_2 = 1/(8\pi) \approx 0.04$ (the $\sim 25\times$ suppression).
6. `test_upper_critical_dim_C2_C3` — $d_c(2) = 4$, $d_c(3) = 3$ from `upper_critical_dim_for_cusp`.

7. `test_C3_skeleton_is_4_3` — $1 + 1/3 = 4/3$.
8. `test_observed_tau_close_to_C3_skeleton` — Manna $\tau \approx 1.275$ within 5% of 4/3; C-DP $\tau \approx 1.338$ within 1% of 4/3.
9. `test_dressing_scale_consistent_with_observation` — $B_3 \cdot \varepsilon^2$ with class counting in the range ~ 5–200 matches the observed gap $\Delta\tau \approx 0.06$.

7 Directions we believe do *not* currently slot

Scouting four further problems from the CFAC candidate list (Amarteifio, 2026b, `docs/problems.md`) returned clear structural reasons why the current stratification framework does not reach them. These are not proven no-gos: each flags a specific extension that would be required. We state them here to mark the boundary of the framework honestly.

7.1 LERW in $d = 3$

Loop-erased random walks in $d = 3$ have a scaling exponent $\nu \approx 1.624$ that has resisted closed form since Kozma (Kozma, 2007) proved it is not equal to 3/2. Wilson's algorithm connects LERW to spanning-tree enumeration, i.e. to the Kirchhoff polynomial that CFAC computes natively. However:

- In the thermodynamic limit the LERW generating function on $\mathbb{Z}^3$ is a lattice Green's function, not algebraic and not D-finite; Theorem 1 does not apply.
- Setting $\tau = \nu$ would require $k = 1/(\nu - 1) \approx 1.60$, neither integer nor dyadic.

We believe LERW 3D is not reachable without extending the stratification framework to non-D-finite generating functions. A legitimate adjacent calculation would be the tropical Kirchhoff decomposition of Wilson-weight expectations on *finite* graphs.

7.2 TAP complexity in spin glasses

The log-count of critical points of a random spin-glass energy landscape — the Bray–Moore / Crisanti–Leuzzi complexity (Bray and Moore, 1980; Crisanti and Leuzzi, 2005), sharpened by Auffinger–Ben Arous–Černý (Auffinger, Ben Arous and Černý, 2013) and Subag (Subag, 2017) for pure p-spin — is controlled by the Kac–Rice formula $\mathbb{E}|\det(H - fI)|$, where H is the Hessian, an $N \times N$ GOE matrix. The CFAC bridge function (Amarteifio, 2026a, Section 4) is scalar: it handles mass-independent scalar self-energies and rank-k polynomial integrands. The object that would describe TAP complexity is a *matrix-valued* bridge — the operator generalisation whose scalar limit recovers Eq. (3) but whose determinantal structure captures the GOE spectral measure. We believe this is a genuine extension requiring new mathematics (free probability, R-transforms), not a mechanical tensor dressing of the existing bridge. TAP complexity is parked pending that extension.

7.3 KPZ upper-tail universality

The KPZ equation in 1+1d has an upper-tail rate function of shape $|h|^{3/2}$ with a coefficient known only for integrable weight distributions (Corwin and Ghosal, 2020; Krajenbrink, 2020). Universality beyond integrable weights is one of the top open problems in integrable probability.

- Structural match: $\tau = 3/2$ is exactly the $\mathcal{C}_2$ skeleton (square-root branch), the generic Puiseux class of a polynomial DSE. The tilted-DSE machinery in `tilted.py` handles SCGFs of polynomial DSEs. Reproducing the *exponent* is, structurally, cheap.
- Missing machinery: the *coefficient* requires a bridge function for disorder-averaged random-weight transfer matrices. No such bridge is currently implemented; the existing bridges cover deterministic Doi–Peliti + MSR only.

We believe KPZ upper-tail universality is a "cautious go" for the scoping layer (show that $\tau = 3/2$ comes out of the tilted directed-polymer DSE at the $\mathcal{C}_2$ stratum) and a research programme for the full universality proof.

7.4 Why listing these matters

The stratification theorem, as stated, is narrow: it applies to polynomial DSEs whose generating functions are algebraic or D-finite. Each negative case above identifies a specific structural feature (non-D-finite GF, matrix-valued bridge, disorder-averaged bridge) that the current framework does not handle. We list them here rather than in a separate note because the positive result (Manna/C-DP on $\mathcal{C}_3$) is only interpretable against the clear boundary of what the framework currently covers.

8 Universal scope: CRN-in-field as the underlying object

We have framed the Manna calculation in narrow terms (one universality class, one stratum), but the machinery used here is not Manna-specific. This section makes the wider scope claim explicit so that the Manna result can be read as one specific test of it.

The scope claim. *Any statistical-physics system whose dynamics admits a Doi–Peliti / MSR action representation — a chemical reaction network embedded in a (possibly non-equilibrium) field, with retarded propagators where appropriate — falls within the CFAC analytic framework.* The two-step procedure is: **(i)** write the CRN-in-field representation of the system; **(ii)** run the CFAC pipeline (filter diagrams by causal contraction and bridge factorisation, compute Symanzik polynomials, apply BPHZ multiplicities $K_G|_{\alpha=1}$, sector-decompose, and extract the residual list of free integrals).

Demonstrated reach. The CFAC pipeline as it stands has been exercised across a heterogeneous list of systems:

- **LDWC depinning**: 2-loop FRG for an elastic manifold, multi-field MSR action, doubled-propagator hat diagram. (Amarteifio, 2026c)
- **LDWC long-range elasticity**: same pipeline, fractional propagator exponents.
- **LDWC periodic CDW**: same pipeline, $N_{\mathrm{res}} = 0$.
- **Wilson–Fisher O(N) η at 2-loop**: equilibrium ϕ^4, no FRG. Reproduces $\eta = (N{+}2)/[2(N{+}8)^2]\varepsilon^2$.
- **Directed percolation 2-loop**: pure Doi–Peliti. (Amarteifio, 2026d)
- **Manna / C-DP**: this paper (multi-field, conserved, non-equilibrium).

None of these are in the same physics class. What they share is the CRN-in-field structure. This is evidence for the scope claim, not proof.

Routes from common stat-phys to CRN-in-field.

Reaction–diffusion: Direct (Doi–Peliti).

Equilibrium ϕ^4 / O(N): Stochastic quantisation (Parisi–Wu) gives a Langevin equation; its MSR action is a CRN-in-field. The field-theoretic anomalous dimension η falls out of the same pipeline as Manna's τ.

Spin systems: Glauber dynamics is literally a CRN of spin flips; mapping to a continuous order parameter is standard.

KPZ / surface growth: MSR action for the height field; the noise vertex defines a CRN.

Hydrodynamic fluctuations: MSR for Navier–Stokes; the pressure constraint is a Lagrange-multiplier field; the noise is a CRN.

Lattice gauge (stochastic quantisation): Langevin in field space, MSR action with gauge-fixing constraints.

The technical step in each case is well-known; CFAC inherits the machinery once the action is in standard form.

Where the scope claim genuinely doesn't reach (yet). The boundary cases of Section 7 sharpen this:

- **Matrix-valued fields** (TAP complexity, RMT, replica $n \to 0$): the CRN is matrix-valued, and the bridge function in CFAC is currently scalar. A matrix-valued bridge (free probability / R-transforms) is a research extension.
- **Non-D-finite generating functions** (LERW in $d = 3$, lattice Green's functions): the stratification theorem requires algebraic or D-finite GFs.
- **Topologically constrained systems** (BRST-quantised gauge theory, field theories with global anomalies): the constraint algebra is not currently handled by the pipeline.

Manna's role in the wider programme. Manna is deliberately a stress test for the multi-field, conserved, non-equilibrium corner of the scope. If CFAC handles Manna successfully (the slotting holds, the rank-3 bridge gives the right order of magnitude, and the conservation projector in the multi-field DSE selects $\mathcal{C}_3$ as predicted), it is direct evidence that the wider scope claim is plausible: *a representative system from a distinct stat-phys corner falls in line with the same machinery.* A separate note in preparation will state the scope claim formally and tabulate the demonstrated cases as a reference artefact.

9 Discussion

If the γ_3 computation in Section 4 reproduces the C-DP/Manna $\tau \approx 1.30$ within simulation error bars, the stratification theorem provides a genuinely new organising principle for non-DP universality: classes are grouped by their skeleton k, with observable τ differences arising from calculable dressings rather than being independent phenomenological parameters.

What is *not* claimed. We do not claim to prove the 1+1d Manna exponent analytically. We claim that the *slotting* — the organisation of the universality landscape by $\mathcal{C}_k$ — is a correct coarse structure, falsifiable by the γ_3 prediction. A failure of that prediction would rule out the slotting.

Next steps. (i) Complete the $\gamma_3(d = 1 + 1)$ calculation in Section 4 and confirm or falsify the τ-gap prediction. (ii) Extend to the voter class at $k \to \infty$ with log corrections. (iii) Test the slotting on a further non-DP class not used in the hypothesis formation (parity-conserving branching annihilation random walks, for instance).

A Related coupled-field systems: ants (TODO)

A natural sibling of CDP/Manna in the CFAC programme is the chemotactic ant colony, treated in (Amarteifio, 2026e) as the canonical coupled-field system: a Doi–Peliti particle sector (ants in two states $S \rightleftharpoons R$) coupled to an MSR continuous-field sector (pheromone density). The ant construction shares the multi-field, non-equilibrium structure that made CDP/Manna the test case in this paper.

The two systems sit on *different* Puiseux strata, however:

- CDP/Manna: $\mathcal{C}_3$ (skeleton $\tau_0 = 4/3$, this paper).
- Ants: $\mathcal{C}_2$ saddle-node, with order parameter exponent $\beta = 1/2$ verified to 2.7% by simulation ((Amarteifio, 2026e, §4)). Conservation enters differently — the ant "conservation" is the total particle number, which fixes $\langle S{+}R \rangle$ but does not generate the same Ward identity that promotes CDP from $\mathcal{C}_2$ to $\mathcal{C}_3$.

The two cases together would be a stronger test of the codimension argument (Proposition 3) than either alone:

- **TODO 1.** Apply the `conserved_2.py` (and eventual `conserved_3.py`) machinery to the ant DSE and verify that it returns $\mathcal{C}_2$ dominant, with $\beta = 1/2$ falling out of the rank-2 bridge as in standard DP.
- **TODO 2.** Identify the precise reason the ant conservation does *not* promote to $\mathcal{C}_3$: is it that the ant's $S + R$ conservation is an exact lattice constraint rather than a Ward identity at the field-theoretic level? Or that the conservation acts on the activity sector itself rather than coupling activity to a separate conserved background?
- **TODO 3.** Construct a modified ant model that adds a *second* conservation law (e.g., a conserved environmental field) and check whether the codimension argument predicts $\mathcal{C}_4$ promotion, observing what numerical β emerges.

These cross-tests would either strengthen the slotting framework (by showing it correctly predicts *which* coupled-field systems sit on which strata) or expose its failure mode (a coupled-field system with conservation that nonetheless does not slot as predicted). Both outcomes are informative. We defer this work to a follow-up note.

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